

**This assignment will not be graded and is only for practice.**

### Level 1

1) Suppose  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is a linear transformation, such that  $T(x, y) = (x, 0)$ . Which of the following options are correct? **1 point**

- ☐ The matrix corresponding to  $T$  with respect to the standard ordered basis of  $\mathbb{R}^2$ , i.e.,  $\{(1, 0), (0, 1)\}$ , for both the domain and co-domain is  $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- ☐ The matrix corresponding to  $T$  with respect to the standard ordered basis of  $\mathbb{R}^2$ , i.e.,  $\{(1, 0), (0, 1)\}$ , for both the domain and co-domain is  $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ .
- ☐  $T$  is neither one-one nor onto.
- ☐  $T$  is one-one but not onto.

No, the answer is incorrect.

Score: 0

### Accepted Answers:

The matrix corresponding to  $T$  with respect to the standard ordered basis of  $\mathbb{R}^2$ , i.e.,  $\{(1, 0), (0, 1)\}$ , for both the domain and co-domain is  $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ .

$T$  is neither one-one nor onto.

2) If the matrix corresponding to a linear transformation  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ , with respect to standard ordered basis of  $\mathbb{R}^2$ , i.e.,  $\{(1, 0), (0, 1)\}$ , for both the domain and co-domain, is  $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$ . Which of the following is the appropriate definition of  $T$ ? **1 point**

- ☐  $T(x, y) = (2x + y, 3x + 4y)$
- ☐  $T(x, y) = (2x + y, 3x - 4y)$
- ☐  $T(x, y) = (x + 4y, 2x + 3y)$
- ☐  $T(x, y) = (2x + 3y, x + 4y)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$T(x, y) = (2x + 3y, x + 4y)$$

Let  $W = \{(x, y, z) \mid x = 2y + z\}$  be a subspace of  $\mathbb{R}^3$ . Let  $\beta = \{(2, 1, 0), (1, 0, 1)\}$  be a basis of  $W$ . Let  $T : W \rightarrow \mathbb{R}^2$  be a linear transformation such that  $T(2, 1, 0) = (1, 0)$  and  $T(1, 0, 1) = (0, 1)$ . Answer the questions 3, 4 and 5 using the given information.

3) Which of the following is the appropriate definition of  $T$ ?

**1 point**

- ☐  $T(x, y, z) = (x, y)$
- ☐  $T(x, y, z) = (x, z)$
- ☐  $T(x, y, z) = (y, z)$
- ☐  $T(x, y, z) = (x - y - z, x - 2y)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$T(x, y, z) = (y, z)$$

$$T(x, y, z) = (x - y - z, x - 2y)$$

4) Choose the correct options.

**1 point**

- ☐  $T$  is one to one but not onto.
- ☐  $T$  is onto but not one to one.
- ☐  $T$  is neither one to one nor onto.
- ☐  $T$  is an isomorphism.

No, the answer is incorrect.

Score: 0

## Accepted Answers:

 $T$  is an isomorphism.

5) What will be the matrix representation of  $T$  with respect to the basis  $\beta$  for  $W$  and  $\gamma = \{(1, 1), (1, -1)\}$  for  $\mathbb{R}^2$ ?

**1 point**

☐  $\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

☐  $\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

☐  $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

☐  $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

## Accepted Answers:

$$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

6) Choose the set of correct options.

**1 point**

☐ Let  $u = (3, 1, 0)$ ,  $v = (0, 1, 7)$ , and  $w = (3, 0, -7)$ . There is a linear transformation  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  such that  $T(u) = T(v) = (0, 0, 0)$  and that  $T(w) = (5, 1, 0)$ .

☐ Let  $u = (2, 1, 0)$ ,  $v = (1, 0, 1)$ , and  $w = (3, 5, 6)$ . There is a linear transformation  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$  such that  $T(u) = (1, 0)$ ,  $T(v) = (0, 1)$  and that  $T(w) = (5, 6)$ .

☐ Let  $u = (1, 0, 0)$ ,  $v = (1, 0, 1)$ , and  $w = (0, 1, 0)$ . There is a linear transformation  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  such that  $T(u) = (0, 0, 0)$ ,  $T(v) = (0, 0, 0)$ ,  $T(w) = (0, 0, 0)$ , and  $T(1, 4, 2) = (2, 4, 1)$ .

☐ Let  $u = (1, 0)$ , and  $v = (0, 1)$ . There is a linear transformation  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  such that  $T(u) = (\pi, 1)$ , and  $T(v) = (1, e)$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let  $u = (2, 1, 0)$ ,  $v = (1, 0, 1)$ , and  $w = (3, 5, 6)$ . There is a linear transformation  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$  such that  $T(u) = (1, 0)$ ,  $T(v) = (0, 1)$  and that  $T(w) = (5, 6)$ .

Let  $u = (1, 0)$ , and  $v = (0, 1)$ . There is a linear transformation  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  such that  $T(u) = (\pi, 1)$ , and  $T(v) = (1, e)$ .

## Level 2

7) Choose the set of correct options.

1 point

- ☐ If the Identity matrix of order 2 is the matrix representation of a linear transformation  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ , with respect to the standard ordered basis for both the domain and co-domain, then it is also the matrix representation of  $T$  with respect to any other basis  $\beta$  for both the domain and co-domain.
- ☐ Let  $T$  be an isomorphism between two vector spaces, and  $A$  be the matrix representation of  $T$  with respect to some basis. Then the nullity of  $A$  is 0.
- ☐ If  $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$  is the matrix representation of  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  with respect to the standard ordered basis of  $\mathbb{R}^2$ , for both the domain and co-domain, then the matrix representation of  $T \circ T$  with respect to the same bases will also be  $A$ .
- ☐ If  $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$  is the matrix representation of  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  with respect to the standard ordered basis of  $\mathbb{R}^2$ , for both the domain and co-domain, then the matrix representation of  $T \circ T$  will be the identity matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If the Identity matrix of order 2 is the matrix representation of a linear transformation  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ , with respect to the standard ordered basis for both the domain and co-domain, then it is also the matrix representation of  $T$  with respect to any other basis  $\beta$  for both the domain and co-domain.

Let  $T$  be an isomorphism between two vector spaces, and  $A$  be the matrix representation of  $T$  with

respect to some basis. Then the nullity of  $A$  is 0.

If  $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$  is the matrix representation of  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  with respect to the standard ordered basis of  $\mathbb{R}^2$ , for both the domain and co-domain, then the matrix representation of  $T \circ T$  with respect to the same bases will also be  $A$ .

Let  $\beta = \{(1, 0), (0, 1)\}$  and  $\gamma = \{(1, 1), (1, -1)\}$  be two bases of  $\mathbb{R}^2$ . If  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$  is the matrix representation of a linear transformation  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  with respect to  $\beta$  for both the domain and the co-domain. Answer questions 8 and 9 using the information given above.

8) What is the matrix representation of  $T$  with respect to  $\gamma$  for the domain and  $\beta$  for the co-domain?

**1 point**

- ☐  $\begin{bmatrix} a+b & c+d \\ a-b & c-d \end{bmatrix}$
- ☐  $\begin{bmatrix} a & -b \\ c & -d \end{bmatrix}$
- ☐  $\begin{bmatrix} a+b & a-b \\ c+d & c-d \end{bmatrix}$
- ☐  $\begin{bmatrix} a+b & -a-b \\ c+d & -c-d \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} a+b & a-b \\ c+d & c-d \end{bmatrix}$$

9) What is the matrix representation of  $T$  with respect to  $\gamma$  for both the domain and the co-domain?

**1 point**

- ☐  $\begin{bmatrix} a+b+c+d & a-b+c-d \\ a+b-c-d & a-b-c+d \end{bmatrix}$
- ☐  $\begin{bmatrix} \frac{a+b+c+d}{2} & \frac{a-b+c-d}{2} \\ \frac{a+b-c-d}{2} & \frac{a-b-c+d}{2} \end{bmatrix}$

- ☐  $\begin{bmatrix} \frac{a+b+c+d}{2} & \frac{a+b-c-d}{2} \\ \frac{a-b+c-d}{2} & \frac{a-b-c+d}{2} \end{bmatrix}$
- ☐  $\begin{bmatrix} a+b+c+d & a+b-c-d \\ a-b+c-d & a-b-c+d \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} \frac{a+b+c+d}{2} & \frac{a-b+c-d}{2} \\ \frac{a-b-c+d}{2} & \frac{a-b-c+d}{2} \end{bmatrix}$$

10) Which of the following options are correct?

**1 point**

- ☐ If  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is a linear transformation which is both one-one and onto then  $T^{-1}$  is also a linear transformation.
- ☐ If  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  then the matrix corresponding to  $T$ , with respect to the standard bases for  $\mathbb{R}^n$  and  $\mathbb{R}^m$  is of order  $n \times m$ .
- ☐ If  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  then the matrix corresponding to  $T$ , with respect to the standard bases for  $\mathbb{R}^n$  and  $\mathbb{R}^m$  is of order  $m \times n$ .
- ☐ Consider the vector space  $V = \left\{ \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \mid a, b, c \in \mathbb{R}, \right\}$ , that is, the set of all  $2 \times 2$  upper triangular matrices with usual matrix addition and scalar multiplication. Then there exists an isomorphism from  $\mathbb{R}^3$  to  $V$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

If  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is a linear transformation which is both one-one and onto then  $T^{-1}$  is also a linear transformation.

If  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  then the matrix corresponding to  $T$ , with respect to the standard bases for  $\mathbb{R}^n$  and  $\mathbb{R}^m$  is of order  $m \times n$ .

Consider the vector space  $V = \left\{ \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \mid a, b, c \in \mathbb{R}, \right\}$ , that is, the set of all  $2 \times 2$  upper triangular matrices with usual matrix addition and scalar multiplication. Then there exists an isomorphism from  $\mathbb{R}^3$  to  $V$ .

**Your score is: 0/10**